



gemma-7b-it

Beta

Text Generation • Google

`@hf/google/gemma-7b-it`

Gemma is a family of lightweight, state-of-the-art open models from Google, built from the same research and technology used to create the Gemini models. They are text-to-text, decoder-only large language models, available in English, with open weights, pre-trained variants, and instruction-tuned variants.

Model Info

Context Window [↗](#)

8,192 tokens

Terms and License

[link ↗](#)

More information

[link ↗](#)

Model Info

LoRA	Yes
Beta	Yes

Playground

Try out this model with Workers AI LLM Playground. It does not require any setup or authentication and an instant way to preview and test a model directly in the browser.

Launch the LLM Playground

Usage

✓ Worker - Streaming

TypeScript

```
export interface Env {  
  AI: Ai;  
}  
  
export default {  
  async fetch(request, env):  
    Promise<Response> {  
  
    const messages = [  
      { role: "system", content: "You are a  
friendly assistant" },  
      {  
        role: "user",  
        content: "What is the origin of the  
phrase Hello, World",  
      },  
    ];  
  
    const stream = await  
env.AI.run("@hf/google/gemma-7b-it", {  
      messages,  
      stream: true,  
    });
```

```
return new Response(stream, {  
  headers: { "content-type":  
"text/event-stream" },  
  });  
},  
} satisfies ExportedHandler<Env>;
```

▼ Worker

TypeScript

```
export interface Env {  
  AI: Ai;  
}  
  
export default {  
  async fetch(request, env):  
    Promise<Response> {  
  
    const messages = [  
      { role: "system", content: "You are a  
friendly assistant" },  
      {  
        role: "user",  
        content: "What is the origin of the  
phrase Hello, World",  
      },  
    ];  
  
    const response = await  
env.AI.run("@hf/google/gemma-7b-it", {  
  messages });  
  
    return Response.json(response);  
  },  
} satisfies ExportedHandler<Env>;
```

▼ Python

Python

```
import os
import requests

ACCOUNT_ID = "your-account-id"
AUTH_TOKEN = os.environ.get("CLOUDFLARE_AUTH_TOKEN")

prompt = "Tell me all about PEP-8"
response = requests.post(
    f"https://api.cloudflare.com/client/v4/accounts/{ACCOUNT_ID}/workers-ai/generate",
    headers={"Authorization": f"Bearer {AUTH_TOKEN}"},
    json={
        "messages": [
            {"role": "system", "content": "You are a helpful assistant."},
            {"role": "user", "content": prompt}
        ]
    }
)
result = response.json()
print(result)
```

▼ curl

```
curl
```

```
https://api.cloudflare.com/client/v4/accounts/  
7b-it \  
-X POST \  
-H "Authorization: Bearer $CLOUDFLARE_AUTH_TOKEN" \  
-d '{ "messages": [{ "role": "system", "content": "You are a helpful assistant." },  
"user", "content": "Why is pizza so good" } ] }'
```

① OpenAI compatible endpoints

Workers AI also supports OpenAI compatible API endpoints for `/v1/chat/completions` and `/v1/embeddings`. For more details, refer to [Configurations](#).

Parameters

* indicates a required field

Input

• **object**

- **prompt** **string** required min 1

The input text prompt for the model to generate a response.

- **lora** **string**

Name of the LoRA (Low-Rank Adaptation) model to fine-tune the base model.

- **response_format** **object**

- **type** **string**

- **json_schema**

- **raw** **boolean**

If true, a chat template is not applied and you must adhere to the specific model's expected formatting.

- **stream** **boolean**

If true, the response will be streamed back incrementally using SSE, Server Sent Events.

- `max_tokens` **integer** default 256

The maximum number of tokens to generate in the response.

- `temperature` **number** default 0.6 min 0 max 5

Controls the randomness of the output; higher values produce more random results.

- `top_p` **number** min 0.001 max 1

Adjusts the creativity of the AI's responses by controlling how many possible words it considers. Lower values make outputs more predictable; higher values allow for more varied and creative responses.

- `top_k` **integer** min 1 max 50

Limits the AI to choose from the top 'k' most probable words. Lower values make responses more focused; higher values introduce more variety and potential surprises.

- **seed** **integer** min 1 max 9999999999

Random seed for reproducibility of the generation.

- **repetition_penalty** **number** min 0 max 2

Penalty for repeated tokens; higher values discourage repetition.

- **frequency_penalty** **number** min -2 max 2

Decreases the likelihood of the model repeating the same lines verbatim.

- **presence_penalty** **number** min -2 max 2

Increases the likelihood of the model introducing new topics.

- ▶ **1** **object**

- **messages** **array** required

An array of message objects representing the conversation history.

- **items** **object**

- **role** **string** required

The role of the message sender (e.g., 'user', 'assistant', 'system', 'tool').

- **content** **string** required

The content of the message as a string.

- **functions** **array**

- **items** **object**

- **name** **string** required

- **code** **string** required

- **tools** **array**

A list of tools available for the assistant to use.

- **items** **one of**

- **0** **object**

- **name** **string** required

The name of the tool. More descriptive the better.

- **description** **string** required

A brief description of what the tool does.

- **parameters** **object** required

Schema defining the parameters accepted by the tool.

- **type** **string** required

The type of the parameters object (usually 'object').

- **required** **array**

List of required parameter names.

- **items** **string**

- **properties** **object** required

Definitions of each parameter.

- **additionalProperties** **object**

- **type** **string** required

The data type of the parameter.

- **description** **string** required

A description of the expected parameter.

■ 1 **object**

- **type** **string** required

Specifies the type of tool (e.g., 'function').

- **function** **object** required

Details of the function tool.

- **name** **string** required

The name of the function.

- **description** **string** required

A brief description of what the function does.

- **parameters** **object** required

Schema defining the parameters accepted by the function.

- **type** **string** required

The type of the parameters object (usually 'object').

- **required** **array**

List of required parameter names.

- **items** **string**

- **properties** **object** required

Definitions of each parameter.

- **additionalProperties** **object**

- **type** **string** required

The data type of the parameter.

- **description** **string** required

A description of the expected parameter.

- **response_format** **object**

- **type** **string**

- **json_schema**

- **raw** **boolean**

If true, a chat template is not applied and you must adhere to the specific model's expected formatting.

- **stream** **boolean**

If true, the response will be streamed back incrementally using SSE, Server Sent Events.

- **max_tokens** **integer** default 256

The maximum number of tokens to generate in the response.

- **temperature** **number** default 0.6 min 0 max 5

Controls the randomness of the output; higher values produce more random results.

- **top_p** **number** min 0.001 max 1

Adjusts the creativity of the AI's responses by controlling how many possible words it considers. Lower values make outputs more predictable; higher values allow for more varied and creative responses.

- `top_k` **integer** min 1 max 50

Limits the AI to choose from the top 'k' most probable words. Lower values make responses more focused; higher values introduce more variety and potential surprises.

- `seed` **integer** min 1 max 9999999999

Random seed for reproducibility of the generation.

- `repetition_penalty` **number** min 0 max 2

Penalty for repeated tokens; higher values discourage repetition.

- `frequency_penalty` **number** min -2 max 2

Decreases the likelihood of the model repeating the same lines verbatim.

- `presence_penalty` **number** min -2 max 2

Increases the likelihood of the model introducing new topics.

Output

• **0** **object**

- response **string** required

The generated text response from the model

- usage **object**

Usage statistics for the inference request

- prompt_tokens **number** 0

Total number of tokens in input

- completion_tokens **number** 0

Total number of tokens in output

- total_tokens **number** 0

Total number of input and output tokens

- tool_calls **array**

An array of tool calls requests made during the response generation

- items **object**

- arguments **object**

The arguments passed to be passed to the tool call request

- name **string**

The name of the tool to be called

- 1 **string**

API Schemas

The following schemas are based on JSON Schema

Input

Output

```
{  
  "type": "object",  
  "oneOf": [  
    {  
      "title": "Prompt",  
      "properties": {  
        "prompt": {  
          "type": "string",  
          "minLength": 1,  
          "description": "The input  
text prompt for the model to generate a  
response."  
        },  
        "lora": {  
          "type": "string",  
          "description": "Name of  
the LoRA (Low-Rank Adaptation) model to fine-  
tune the base model."  
        },  
        "response_format": {  
          "title": "JSON Mode",  
          "type": "object",  
          "properties": {  
            "type": {  

```

```
        "type": "string",
        "enum": [
            "json_object",
            "json_schema"
        ]
    },
    "json_schema": {}
},
"raw": {
    "type": "boolean",
    "default": false,
    "description": "If true, a
chat template is not applied and you must
adhere to the specific model's expected
formatting."
},
"stream": {
    "type": "boolean",
    "default": false,
    "description": "If true,
the response will be streamed back
incrementally using SSE, Server Sent Events."
},
"max_tokens": {
```



```
    "type": "integer",
    "default": 256,
    "description": "The
maximum number of tokens to generate in the
response."
  },
  "temperature": {
    "type": "number",
    "default": 0.6,
    "minimum": 0,
    "maximum": 5,
    "description": "Controls
the randomness of the output; higher values
produce more random results."
  },
  "top_p": {
    "type": "number",
    "minimum": 0.001,
    "maximum": 1,
    "description": "Adjusts
the creativity of the AI's responses by
controlling how many possible words it
considers. Lower values make outputs more
predictable; higher values allow for more
varied and creative responses."
```

```
    },  
    "top_k": {  
      "type": "integer",  
      "minimum": 1,  
      "maximum": 50,  
      "description": "Limits the  
AI to choose from the top 'k' most probable  
words. Lower values make responses more  
focused; higher values introduce more variety  
and potential surprises."  
    },  
    "seed": {  
      "type": "integer",  
      "minimum": 1,  
      "maximum": 9999999999,  
      "description": "Random  
seed for reproducibility of the generation."  
    },  
    "repetition_penalty": {  
      "type": "number",  
      "minimum": 0,  
      "maximum": 2,  
      "description": "Penalty  
for repeated tokens; higher values discourage  
repetition."
```

```
    },  
    "frequency_penalty": {  
      "type": "number",  
      "minimum": -2,  
      "maximum": 2,  
      "description": "Decreases  
the likelihood of the model repeating the same  
lines verbatim."  
    },  
    "presence_penalty": {  
      "type": "number",  
      "minimum": -2,  
      "maximum": 2,  
      "description": "Increases  
the likelihood of the model introducing new  
topics."  
    }  
  },  
  "required": [  
    "prompt"  
  ]  
},  
{  
  "title": "Messages",  
  "properties": {
```

```
"messages": {  
  "type": "array",  
  "description": "An array  
of message objects representing the  
conversation history.",  
  "items": {  
    "type": "object",  
    "properties": {  
      "role": {  
        "type":  
"string",  
        "description":  
"The role of the message sender (e.g., 'user',  
'assistant', 'system', 'tool')."  
      },  
      "content": {  
        "type":  
"string",  
        "description":  
"The content of the message as a string."  
      }  
    },  
    "required": [  
      "role",  
      "content"
```

```
    ]
  }
},
"functions": {
  "type": "array",
  "items": {
    "type": "object",
    "properties": {
      "name": {
        "type":
"string"

      },
      "code": {
        "type":
"string"

      }
    },
    "required": [
      "name",
      "code"
    ]
  }
},
"tools": {
  "type": "array",
```

```
"description": "A list of
tools available for the assistant to use.",
"items": {
  "type": "object",
  "oneOf": [
    {
      "properties":
{
      "name": {
        "type": "string",

"description": "The name of the tool. More
descriptive the better."
      },

"description": {
        "type": "string",

"description": "A brief description of what
the tool does."
      },

"parameters": {
```

```
"type": "object",

"description": "Schema defining the parameters
accepted by the tool.",

"properties": {

  "type": {

    "type": "string",

    "description": "The type of the parameters
object (usually 'object')."

    },

    "required": {

      "type": "array",

      "description": "List of required parameter
names.",

      "items": {
```

```
"type": "string"

},

"properties": {

  "type": "object",

  "description": "Definitions of each
parameter.",

  "additionalProperties": {

    "type": "object",

    "properties": {

      "type": {

        "type": "string",

        "description": "The data type of the
parameter."
```



```
},  
  
"description": {  
  
  "type": "string",  
  
  "description": "A description of the expected  
parameter."  
  
}  
  
},  
  
"required": [  
  
  "type",  
  
  "description"  
  
]  
  
}  
  
},
```

```
"required": [  
  
  "type",  
  
  "properties"  
  
],  
},  
"required": [  
  "name",  
  
  "description",  
  
  "parameters"  
],  
},  
{  
  "properties":  
  
    "type": {  
  
      "type": "string",  
  
      "description": "Specifies the type of tool"
```

```
(e.g., 'function')."
```

```
},
```

```
"function": {
```

```
"type": "object",
```

```
"description": "Details of the function  
tool.",
```

```
"properties": {
```

```
"name": {
```

```
"type": "string",
```

```
"description": "The name of the function."
```

```
},
```

```
"description": {
```

```
"type": "string",
```

```
"description": "A brief description of what  
the function does."
```

},

```
"parameters": {  
  
  "type": "object",  
  
  "description": "Schema defining the parameters  
accepted by the function.",  
  
  "properties": {  
  
    "type": {  
  
      "type": "string",  
  
      "description": "The type of the parameters  
object (usually 'object')."  
  
    },  
  
    "required": {  
  
      "type": "array",  
  
      "description": "List of required parameter
```

```
names.",  
  
"items": {  
  
"type": "string"  
  
}  
  
},  
  
"properties": {  
  
"type": "object",  
  
"description": "Definitions of each  
parameter.",  
  
"additionalProperties": {  
  
"type": "object",  
  
"properties": {  
  
"type": {
```

```
"type": "string",

"description": "The data type of the
parameter."

},

"description": {

"type": "string",

"description": "A description of the expected
parameter."

}

},

"required": [

"type",

"description"
```

```
]
```

```
}
```

```
}
```

```
},
```

```
"required": [
```

```
"type",
```

```
"properties"
```

```
]
```

```
}
```

```
},
```

```
"required": [
```

```
"name",
```

```
"description",
```

"parameters"

```
    ]
  },
  "required": [
    "type",
    "function"
  ]
}
]
}
},
"response_format": {
  "title": "JSON Mode",
  "type": "object",
  "properties": {
    "type": {
      "type": "string",
      "enum": [
        "json_object",
        "json_schema"
      ]
    },
    "json_schema": {}
  }
}
```



```
    }  
  },  
  "raw": {  
    "type": "boolean",  
    "default": false,  
    "description": "If true, a  
chat template is not applied and you must  
adhere to the specific model's expected  
formatting."  
  },  
  "stream": {  
    "type": "boolean",  
    "default": false,  
    "description": "If true,  
the response will be streamed back  
incrementally using SSE, Server Sent Events."  
  },  
  "max_tokens": {  
    "type": "integer",  
    "default": 256,  
    "description": "The  
maximum number of tokens to generate in the  
response."  
  },  
  "temperature": {
```

```
    "type": "number",
    "default": 0.6,
    "minimum": 0,
    "maximum": 5,
    "description": "Controls
the randomness of the output; higher values
produce more random results."
  },
  "top_p": {
    "type": "number",
    "minimum": 0.001,
    "maximum": 1,
    "description": "Adjusts
the creativity of the AI's responses by
controlling how many possible words it
considers. Lower values make outputs more
predictable; higher values allow for more
varied and creative responses."
  },
  "top_k": {
    "type": "integer",
    "minimum": 1,
    "maximum": 50,
    "description": "Limits the
AI to choose from the top 'k' most probable
```

words. Lower values make responses more focused; higher values introduce more variety and potential surprises."

```
    },  
    "seed": {  
      "type": "integer",  
      "minimum": 1,  
      "maximum": 9999999999,  
      "description": "Random  
seed for reproducibility of the generation."  
    },  
    "repetition_penalty": {  
      "type": "number",  
      "minimum": 0,  
      "maximum": 2,  
      "description": "Penalty  
for repeated tokens; higher values discourage  
repetition."  
    },  
    "frequency_penalty": {  
      "type": "number",  
      "minimum": -2,  
      "maximum": 2,  
      "description": "Decreases  
the likelihood of the model repeating the same
```

```
lines verbatim."
    },
    "presence_penalty": {
      "type": "number",
      "minimum": -2,
      "maximum": 2,
      "description": "Increases
the likelihood of the model introducing new
topics."
    }
  },
  "required": [
    "messages"
  ]
}
]
```

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